

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Solve Any Two of the following.

12

- A) Find vertical deflection at joint C of the truss shown in figure. The area of inclined tie AC is 2000mm^2 while the area of horizontal member BC is 1600mm^2 . Take $E = 200\text{KN/mm}^2$.

(Evaluate)

6

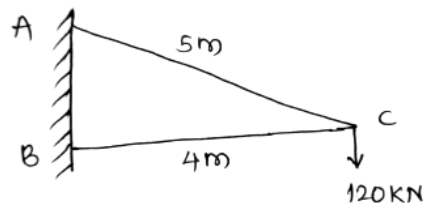


Fig Q.1(a)

- B) A single concentrated load of magnitude W rolls over a simply supported beam of span l . Calculate and draw maximum positive and negative shear force diagrams and maximum bending moment diagram.
- C) Using influence line method find the shear force and bending moment at section K for the beam loaded as shown in figure.

(Understand)

6

(CO3)

6

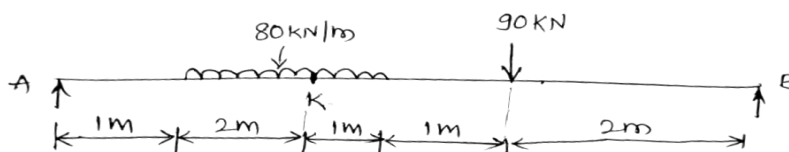


Fig Q.1(c)

Q.2 Solve Any Two of the following.

12

- A) A cable of span l and dip h is subjected to udl w per unit run of horizontal span. If the dip be considered as small as compared with the span. Show that the difference between the greatest and least tension is approximately wh .

(Understand)

6

- B) A three hinged parabolic arch of span 20m, rise 3m carries an udl of 30KN/m on left half of the span. Find the bending moment, radial shear

(Evaluate)

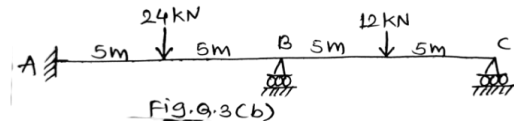
6

force and normal thrust at 6m from left end.

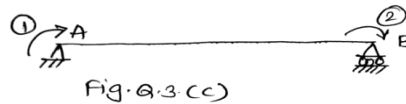
- C) A two hinged parabolic arch of span 20m and rise 4m carries an udl of 20kN/m over entire span. Calculate the horizontal thrust if now a support yields laterally with respect to other by 0.02m, what will be the horizontal thrust? Take $I_0 = 1.7 \times 10^7 \text{ mm}^4$ and $E = 200 \text{ KN/mm}^2$. (Evaluate) 6

Q. 3 Solve Any Two of the following. 12

- A) Derive the relationship between stiffness and flexibility matrix. (CO1) 6
B) Using flexibility matrix method analyse the beam shown in figure. (CO1) 6

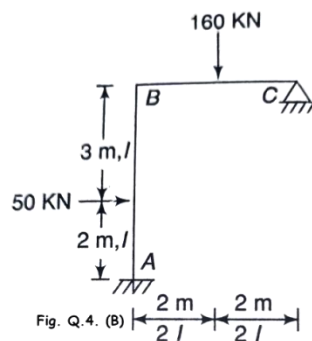


- C) Develop the flexibility matrix for the beam shown below (CO1) 6

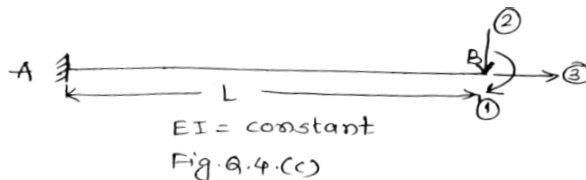


Q.4 Solve Any Two of the following. 12

- A) Differentiate between force method and displacement method. (CO1) 6
B) Analyse the frame shown in fig. using stiffness matrix method. (CO1) 6



- C) Develop the stiffness matrix for the beam shown in figure (CO1) 6



Q. 5 Solve Any Two of the following. 12

- A) Write down general steps of finite element method. (CO2) 6
B) Write a note on Pascal's triangle. (CO2) 6
C) Compute the shape functions for one dimensional element. (CO2) 6

*** End ***